

Fixed-Time Neural Control of Robot Manipulator With Global Stability and Guaranteed Transient Performance

Chengzhi Zhu, Yiming Jiang , and Chenguang Yang , *Senior Member, IEEE*

Abstract—Nowadays, due to many limitations in reality, the optimization of tracking precision and convergence time has attracted the attention of researchers in robotics community. In this article, a fixed-time adaptive neural network (NN) controller is proposed for unknown robot manipulators. A switching mechanism is integrated into the control design such that the semiglobal stability of the conventional NN control systems can be extended to global stability. The time-varying barrier Lyapunov function and the fixed-time control technique are incorporated into the controller design to guarantee the prescribed motion constraints and the fixed-time convergence simultaneously. Compared with some existing fixed-time NN control algorithms, the assumption that NN weight should be upper bounded can be relaxed in our work. Finally, the simulation and experiment studies are respectively carried out based on an unknown 2-degree of freedom robot and a Baxter robot to demonstrate the effectiveness and superiority of the proposed control scheme.

Index Terms—Barrier Lyapunov functions (BLFs), fixed-time convergence, global stability, neural network (NN) control, robot manipulator.

I. INTRODUCTION

NOWADAYS, a variety of effective control schemes [1], [2] have been proposed by researchers in the automation community. In [3], a relatively comprehensive review has been shown to introduce the widely used robot control algorithms.

Manuscript received 12 November 2021; revised 23 January 2022; accepted 14 February 2022. Date of publication 9 March 2022; date of current version 22 August 2022. This work was supported in part by National Natural Science Foundation of China (NSFC) under Grant U20A20200 and Grant 62003136, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2019B1515120076 and Grant 2020B1515120054, and in part by Industrial Key Technologies Research and Development Program of Foshan under Grant 2020001006496 and Grant 2020001006308. (Corresponding author: Chenguang Yang.)

Chengzhi Zhu and Chenguang Yang are with the Key Laboratory of Autonomous Systems and Networked Control, College of Automation Science and Engineering, South China University of Technology, Guangzhou 510640, China (e-mail: auchengzhi@mail.scut.edu.cn; cyang@ieee.org).

Yiming Jiang is with the National Engineering Research Center for Robot Visual Perception and Control, Hunan University, Changsha 410082, China (e-mail: ymjiang@hnu.edu.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIE.2022.3156037>.

Digital Object Identifier 10.1109/TIE.2022.3156037

Most of the proposed control schemes focused on the steady-state tracking accuracy, but few of them involved the transient performance and convergence time.

In practice, the undesirable transient performance may lead to the collision between robots and the surrounding environment, thus causing the system safety problems. Therefore, different controllers have been proposed for robots in the presence of various constraints to achieve desired transient performance [4]–[6]. Recently, the barrier Lyapunov functions (BLFs) were originally proposed to indirectly achieve the state and output constraints in the nonlinear control community [7], [8]. In [7], an adaptive observer-based fuzzy control scheme using BLF has been proposed for a class of systems. In [8], a class of tracking control problems has been investigated for a class of nonlinear systems with deferred full-state constraints.

Due to the complexity of the robot mechanism, the model information about the robots is not available in most practical cases. To solve this problem, the designed controller should compensate the system uncertainty by the input–output data. Neural network (NN) and fuzzy logic technique have been widely employed in intelligent controller design for the unknown systems [6], [9], [10]. However, the above-mentioned NN control schemes can only guarantee semiglobally uniformly ultimately boundedness (SGUUB) of the involved systems, for which the approximation ability of the NN has been greatly attenuated. So it is significant to extend the stability of the NN control schemes from semiglobal stability to global stability. In [11], an adaptive NN controller has been proposed for hypersonic flight vehicle systems to achieve global uniformly ultimately boundedness (GUUB).

Some existing control schemes have been proposed and focused on the convergence time of the systems. However, for the proposed finite-time control algorithms [12], [13], the convergence time is always related to the initial conditions, which are always unavailable in practice. Recently, various fixed-time control algorithms have caught the attention of researchers and been employed in the nonlinear control community [14]–[16]. In [15], the distributed fixed-time algorithm has been proposed for unknown multiple Euler–Lagrange systems under directed graphs. In [16], a fixed-time fuzzy control scheme has been proposed for the tracking problem of a class of nonlinear systems.

So far, there are few researches that combine NN technique and fixed-time control algorithm [17]–[23]. Motivated by [23], a global fixed-time NN controller is proposed for unknown robot

manipulators with predefined output constraints in this article. A BLF has been employed in the controller design such that the tracking errors can be constrained within a predefined range. So the uncertainty of the convergence domain in [17] and [18] can be optimized. Compared with the fixed-time NN control schemes in [17]–[22], a switching mechanism is integrated into the controller design such that all the closed-loop signals can achieve GUUB rather than SGUUB, which can greatly relax the requirement of the NN inputs. Thus, the assumption that the NN weight estimation is bounded in [17]–[19] can be relaxed. So the superiority of the proposed control scheme can be shown. However, compared to our work, the radial basis function neural networks (RBFNNs) in [21]–[23] can achieve less computational complexity because of the superior technique. In our previous work [24], a topology optimization method has been proposed to reduce the data amount for calculation and storage in wireless sensor networks. The above researches are exciting because they may enlighten optimizing the NN structure in our future work. The main contributions of our work are listed as follows.

- 1) A proper BLF has been integrated into the NN controller design such that the prescribed transient performance can be guaranteed and the tracking errors will not violate the output constraints.
- 2) The tracking performance of the controlled robot system can achieve fixed-time convergence regardless of the initial conditions.
- 3) A switching mechanism has been combined with the proposed control scheme to make all the closed-loop signals achieve GUUB such that the assumption that the NN weight estimation is bounded in [17]–[19] can be relaxed.

The rest of this article is organized as follows. The problem formulation and preliminaries which are involved in the control design are introduced in Section II. The global fixed-time NN control design with the guaranteed transient performance and the system stability analysis are described in Section III. The simulation and the experimental studies of the proposed controller are, respectively, presented in Sections IV and V to show the superiority. Finally, Section VI concludes this article.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Problem Formulation

Generally, the dynamic equation of an n -DOF (degree of freedom) robot is described as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau \quad (1)$$

where $q, \dot{q}, \ddot{q} \in R^n$ can be, respectively, expressed as the vectors of joint position, velocity, and acceleration. The terms $M(q), C(q, \dot{q}) \in R^{n \times n}$, and $G(q) \in R^n$, respectively, denote the inertia matrix, Coriolis and centripetal torque matrix, and gravity vector. $\tau \in R^n$ is the control torque applied to the joints.

The objective of the controller design is to make the joint position q in (1) track the desired trajectory q_d with fixed-time convergence regardless of the initial conditions. Meanwhile, the output constraints should not be violated during the control

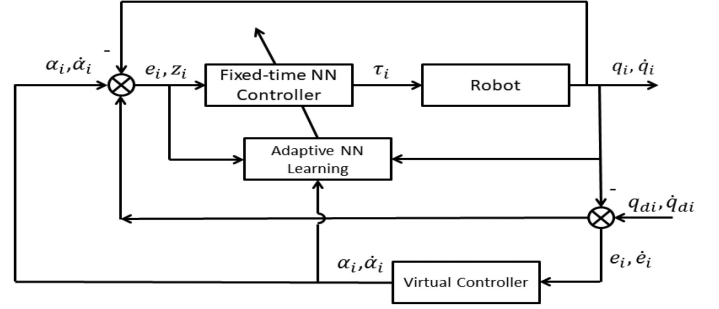


Fig. 1. Framework for the global fixed-time NN robot controller.

operation. Moreover, when the system inputs are not within the NN's approximation domain, the global stability of the closed-loop system should be guaranteed. The controller design strategy is demonstrated in Fig. 1.

B. Preliminaries

Property 1 ([25]): The term $M(q)$ is a symmetric and positive definite matrix. And the term $M(q) - 2C(q, \dot{q})$ is a skew symmetric matrix which satisfies that $\iota^T (\dot{M}(q) - 2C(q, \dot{q})) \iota = 0, \forall \iota \in R^n$.

Assumption 1: The desired trajectory q_d is bounded. The first-order derivative of q_d exists and is continuous. Moreover, $\dot{q}_d$ is also bounded.

Lemma 1 ([26]): For a general Lyapunov function $V(x)$, when the following inequality exists: $\dot{V}(x) \leq -\eta_1 V^{\rho_1}(x) - \beta_1 V^{\rho_2}(x) + a_1$, where a_1 is a positive constant and the parameters $\rho_1 \in (0, 1)$, $\rho_2 \in (1, \infty)$, the system $\dot{x}(t) = f(x)$ can be seen as practical fixed-time stable. And the residual set of the system is

$$\Omega_1 = \left\{ x | V \leq \min \left([a_1 / (\eta_1 (1 - \theta))]^{1/\rho_1}, [a_1 / (\beta_1 (1 - \theta))]^{1/\rho_2} \right) \right\}$$

where θ represents a positive constant satisfying that $\theta \in (0, 1)$. The convergence time of the system is $T \leq T_{\max} = \frac{1}{\eta_1 \theta (1 - \rho_1)} + \frac{1}{\beta_1 \theta (\rho_2 - 1)}$.

Lemma 2 ([27]): The following inequalities hold for any $r_i \in R$ and $p \in R^+$:

$$\sum_{i=1}^n |r_i|^p \geq \left(\sum_{i=1}^n |r_i| \right)^p, 0 < p \leq 1 \quad (2)$$

$$\sum_{i=1}^n |r_i|^p \geq n^{1-p} \left(\sum_{i=1}^n |r_i| \right)^p, p > 1. \quad (3)$$

Lemma 3 ([28]): For any $y \in R$ and $\varpi \in R^+$, the following inequality holds:

$$0 \leq |y| - y \tanh \left(\frac{y}{\varpi} \right) \leq c\varpi \quad (4)$$

where c is a positive constant satisfying that $c = e^{-(c+1)}$, i.e., $c = 0.2785$.

Remark 1 ([29]): A novel function is defined as follows for any $\mu \in R$:

$$\text{sig}^b(\mu) = |\mu|^b \text{sign}(\mu) \quad (5)$$

where b is a positive constant and $\text{sign}(\cdot)$ can be seen as the signum function.

Radial Basis Function Neural Network ([30]): In our work, the RBFNN has been employed to compensate for unknown system function $f(Z) : R^u \rightarrow R$ in the controller design

$$\hat{f}(Z) = \sum_{i=1}^{N_0} \hat{w}_i s_i(Z) = \hat{W}^T S(Z) \quad (6)$$

where $Z \in \Omega_{Z_0} \subset R^u$ and u represent the NN input vector and its dimension, respectively. The term $\hat{W} \in R^{N_0}$ denotes the estimation of the optimal constant weight vector W^* . N_0 denotes the number of the selected NN nodes. The term $S(Z) = [s_1(Z), s_2(Z), \dots, s_{N_0}(Z)]^T \in R^{N_0}$ represents the regression vector, with $s_i(Z)$ being selected as a general radial basis function as follows:

$$s_i(Z, v_i) = \exp \left[\frac{-(Z - v_i)^T (Z - v_i)}{\sigma_i^2} \right] \quad (7)$$

where $v_i \in R^u$ and $\sigma_i \in R$, respectively, denote the center vector and the positive width parameter of the i th NN node. The RBFNN in (6) can approximate the continuous function $f(Z) : R^u \rightarrow R$ to arbitrary accuracy as long as the node number is sufficiently large over the compact set Ω_{Z_0}

$$f(Z) = W^{*T} S(Z) + \varepsilon(Z), \quad \forall Z \in \Omega_{Z_0} \in R^u \quad (8)$$

where $\varepsilon(Z)$ denotes the approximation error of the NN. And $\varepsilon(Z)$ is bounded by a positive constant $\bar{\varepsilon}$, i.e., $|\varepsilon(Z)| \leq \bar{\varepsilon}$ for all $Z \in \Omega_{Z_0}$. The ideal weight vector W^* is the term that minimizes $\varepsilon(Z)$ for all $Z \in \Omega_{Z_0}$, i.e.,

$$W^* = \arg \min_{W \in R^{N_0}} \left\{ \sup_{Z \in \Omega_{Z_0}} |f(Z) - \hat{W}^T S(Z)| \right\}. \quad (9)$$

III. CONTROL DESIGN

Before the controller design, the tracking error signals of the robot manipulator can be defined as follows:

$$e = q - q_d \quad z = \dot{q} - \alpha \quad (10)$$

where $e = [e_1, e_2, \dots, e_n]^T \in R^n$ and $z = [z_1, z_2, \dots, z_n]^T \in R^n$ represent the position tracking error vector and the velocity tracking error vector of the robot manipulator, respectively. The vectors $q \in R^n$ and $q_d \in R^n$ represent the actual trajectory and desired trajectory for the joints, respectively. And $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_n]^T \in R^n$ is the virtual control vector which will be designed later.

A. Requirement for Transient Performance

The boundary functions ξ_{1i} and ξ_{2i} are constructed to constrain the tracking error e_i in different directions as follows:

$$\xi_{1i} = -[(\phi_{0,1i} - \phi_{\infty,1i})e^{-l_{1i}t} + \phi_{\infty,1i}] \quad (11)$$

$$\xi_{2i} = (\phi_{0,2i} - \phi_{\infty,2i})e^{-l_{2i}t} + \phi_{\infty,2i} \quad (12)$$

where $\phi_{0,ji}, \phi_{\infty,ji}, l_{ji}$ ($i = 1, 2, \dots, n, j = 1, 2$) are proper positive constants.

Remark 2: The functions ξ_{2i} and ξ_{1i} are, respectively, defined as the upper and lower bounds of the error e_i where $e_i = q_i - q_{di}$, i.e., the tracking error e_i should be constrained in the range $\xi_{1i} < e_i < \xi_{2i}$. From (11), (12), and the monotonicity of the exponential function, we can specify the transient and steady-state tracking performance by properly selecting the parameters $\phi_{0,ji}, \phi_{\infty,ji}, l_{ji}$ ($i = 1, 2, \dots, n, j = 1, 2$).

B. Global NN Controller Design Involving BLF

The following coordinate transformation is applied on the tracking error vector e :

$$\varphi_a = \left[\frac{e_1}{\xi_{11}}, \frac{e_2}{\xi_{12}}, \dots, \frac{e_n}{\xi_{1n}} \right]^T \quad (13)$$

$$\varphi_b = \left[\frac{e_1}{\xi_{21}}, \frac{e_2}{\xi_{22}}, \dots, \frac{e_n}{\xi_{2n}} \right]^T. \quad (14)$$

Then a new vector φ is defined as $\varphi = [\varphi_1, \varphi_2, \dots, \varphi_n]^T$ where

$$\varphi_i = h_i(e_i)\varphi_{bi} + (1 - h_i(e_i))\varphi_{ai}, \quad i = 1, 2, \dots, n. \quad (15)$$

The terms φ_{ai} and φ_{bi} in (15) are the i th elements of the vectors φ_a and φ_b , respectively. And $h_i(e_i)$ is a function that can be defined as follows:

$$h_i(e_i) = \begin{cases} 1 & e_i \geq 0 \\ 0 & e_i < 0 \end{cases}. \quad (16)$$

Inspired by [31], an asymmetric candidate BLF can be constructed as follows:

$$V_1 = \sum_{i=1}^n \left(\frac{h_i}{2} \ln \frac{1}{1 - \varphi_{bi}^2} + \frac{1 - h_i}{2} \ln \frac{1}{1 - \varphi_{ai}^2} \right) \quad (17)$$

where the initial value of e_i satisfies that $\xi_{1i}(0) < e_i(0) < \xi_{2i}(0)$.

Taking the derivative of (17), we can get

$$\dot{V}_1 = \sum_{i=1}^n \left[h_i \frac{1}{1 - \varphi_{bi}^2} \varphi_{bi} \dot{\varphi}_{bi} + (1 - h_i) \frac{1}{1 - \varphi_{ai}^2} \varphi_{ai} \dot{\varphi}_{ai} \right]. \quad (18)$$

From (13) and (14), we can get that

$$\dot{\varphi}_{ai} = \frac{\dot{e}_i \xi_{1i} - e_i \dot{\xi}_{1i}}{\xi_{1i}^2} \quad \dot{\varphi}_{bi} = \frac{\dot{e}_i \xi_{2i} - e_i \dot{\xi}_{2i}}{\xi_{2i}^2}. \quad (19)$$

Substituting (10) and (19) into (18), we have

$$\begin{aligned} \dot{V}_1 = & \sum_{i=1}^n \left[\frac{h_i \varphi_{bi}}{(1 - \varphi_{bi}^2) \xi_{2i}} \left(z_i + \alpha_i - \dot{q}_{di} - e_i \frac{\dot{\xi}_{2i}}{\xi_{2i}} \right) \right] \\ & + \sum_{i=1}^n \left[\frac{(1 - h_i) \varphi_{ai}}{(1 - \varphi_{ai}^2) \xi_{1i}} \left(z_i + \alpha_i - \dot{q}_{di} - e_i \frac{\dot{\xi}_{1i}}{\xi_{1i}} \right) \right]. \end{aligned} \quad (20)$$

Then the virtual controller α_i can be designed as

$$\begin{aligned} \alpha_i = & \dot{q}_{di} - k_{1i}e_i - \frac{h_i k_{2i} \varphi_{bi}^2 e_i}{1 - \varphi_{bi}^2} - \frac{(1 - h_i) k_{2i} \varphi_{ai}^2 e_i}{1 - \varphi_{ai}^2} \\ & - h_i k_{3i} \left(\frac{1}{2} e_i^2 + \frac{1}{2} \xi_{2i}^2 \right)^{\frac{1}{2}} (1 - \varphi_{bi}^2)^{\frac{1}{4}} - \delta_i e_i \\ & + (1 - h_i) k_{3i} \left(\frac{1}{2} e_i^2 + \frac{1}{2} \xi_{1i}^2 \right)^{\frac{1}{2}} (1 - \varphi_{ai}^2)^{\frac{1}{4}} \end{aligned} \quad (21)$$

where

$$\delta_i = \sqrt{\left(\frac{\dot{\xi}_{1i}}{\xi_{1i}} \right)^2 + \left(\frac{\dot{\xi}_{2i}}{\xi_{2i}} \right)^2} + k_{ai} \quad (22)$$

and k_{1i} , k_{2i} , k_{3i} , and k_{ai} are all proper positive constants. The term k_{ai} is selected to ensure the boundedness of $\dot{\alpha}_i$ when $\dot{\xi}_{1i} = \dot{\xi}_{2i} = 0$.

Substituting (21) into (20), we have

$$\begin{aligned} \dot{V}_1 = & \sum_{i=1}^n \left[\left(\frac{h_i e_i}{\left(1 - \frac{e_i^2}{\xi_{2i}^2}\right) \xi_{2i}^2} + \frac{(1 - h_i) e_i}{\left(1 - \frac{e_i^2}{\xi_{1i}^2}\right) \xi_{1i}^2} \right) z_i \right] \\ & - \sum_{i=1}^n \left[\frac{k_{1i} h_i e_i^2}{\left(1 - \frac{e_i^2}{\xi_{2i}^2}\right) \xi_{2i}^2} + \frac{(1 - h_i) k_{1i} e_i^2}{\left(1 - \frac{e_i^2}{\xi_{1i}^2}\right) \xi_{1i}^2} \right] \\ & - \sum_{i=1}^n \left[\frac{k_{2i} h_i^2 \varphi_{bi}^4}{(1 - \varphi_{bi}^2)^2} + \frac{(1 - h_i)^2 k_{2i} \varphi_{ai}^4}{(1 - \varphi_{ai}^2)^2} \right] \\ & - \sum_{i=1}^n \left[\frac{k_{3i} h_i^2 \varphi_{bi} \sqrt{\frac{1}{2} e_i^2 + \frac{1}{2} \xi_{2i}^2}}{(1 - \varphi_{bi}^2)^{\frac{3}{4}} \xi_{2i}} \right] \\ & + \sum_{i=1}^n \left[\frac{k_{3i} (1 - h_i)^2 \varphi_{ai} \sqrt{\frac{1}{2} e_i^2 + \frac{1}{2} \xi_{1i}^2}}{(1 - \varphi_{ai}^2)^{\frac{3}{4}} \xi_{1i}} \right] \\ & - \sum_{i=1}^n \left[\frac{h_i \varphi_{bi}^2}{(1 - \varphi_{bi}^2)} \left(\delta_i + \frac{\dot{\xi}_{2i}}{\xi_{2i}} \right) + \frac{(1 - h_i) \varphi_{ai}^2}{(1 - \varphi_{ai}^2)} \left(\delta_i + \frac{\dot{\xi}_{1i}}{\xi_{1i}} \right) \right] \end{aligned} \quad (23)$$

where $h_i(1 - h_i) = 0$ has been used for the above derivation.

When $e_i \geq 0$, from (12) and (14), we have

$$- \sum_{i=1}^n \frac{k_{3i} h_i^2 \varphi_{bi} \sqrt{\frac{1}{2} e_i^2 + \frac{1}{2} \xi_{2i}^2}}{(1 - \varphi_{bi}^2)^{\frac{3}{4}} \xi_{2i}} \leq - \sum_{i=1}^n \frac{k_{3i} h_i^2 (\varphi_{bi}^2)^{\frac{3}{4}}}{(1 - \varphi_{bi}^2)^{\frac{3}{4}}} \quad (24)$$

When $e_i < 0$, from (11) and (13), we have

$$\begin{aligned} & \sum_{i=1}^n \frac{k_{3i} (1 - h_i)^2 \varphi_{ai} \sqrt{\frac{1}{2} e_i^2 + \frac{1}{2} \xi_{1i}^2}}{(1 - \varphi_{ai}^2)^{\frac{3}{4}} \xi_{1i}} \\ & \leq - \sum_{i=1}^n \frac{k_{3i} (1 - h_i)^2 (\varphi_{ai}^2)^{\frac{3}{4}}}{(1 - \varphi_{ai}^2)^{\frac{3}{4}}} \end{aligned} \quad (25)$$

By (22), we can obtain the following inequalities:

$$\delta_i + \frac{\dot{\xi}_{2i}}{\xi_{2i}} \geq 0 \quad \delta_i + \frac{\dot{\xi}_{1i}}{\xi_{1i}} \geq 0. \quad (26)$$

Substituting (15), (24), (25), and (26) into (23), we can get that

$$\dot{V}_1 \leq \sum_{i=1}^n \left[- \frac{k_{1i} \varphi_i^2}{1 - \varphi_i^2} - \frac{k_{2i} (\varphi_i^2)^2}{(1 - \varphi_i^2)^2} - \frac{k_{3i} (\varphi_i^2)^{\frac{3}{4}}}{(1 - \varphi_i^2)^{\frac{3}{4}}} + \varsigma_i e_i z_i \right] \quad (27)$$

where

$$\varsigma_i = \frac{h_i}{\xi_{2i}^2 - e_i^2} + \frac{1 - h_i}{\xi_{1i}^2 - e_i^2}. \quad (28)$$

Then a candidate Lyapunov function can be constructed as follows:

$$V_2 = V_1 + \frac{1}{2} z^T M z. \quad (29)$$

Taking the derivative of (29) combined with (1), (10), and Property 1, we have

$$\dot{V}_2 = \dot{V}_1 + z^T (\tau - C\alpha - G - M\dot{\alpha}) \quad (30)$$

where M , C , and G represent the abbreviation of $M(q)$, $C(q, \dot{q})$, and $G(q)$, respectively. The unknown vector which will be approximated by the RBFNNs is defined as

$$F(Z) = -M\dot{\alpha} - C\alpha - G \quad (31)$$

where $F(Z) = [f_1(Z), f_2(Z), \dots, f_n(Z)]^T$. And the input vector can be defined as $Z = [q^T, \dot{q}^T, \alpha^T, \dot{\alpha}^T]^T \in R^{4n}$. It should be noted that the unknown function $f_i(Z)$ approximated by the NN is upper bounded by a known nonnegative smooth function $\bar{f}_i(Z)$ such that $|f_i(Z)| \leq \bar{f}_i(Z) \forall Z \in R^{4n}$.

Substituting (31) into (30), we can obtain that

$$\dot{V}_2 = \dot{V}_1 + z^T [\tau + F(Z)]. \quad (32)$$

Then the RBFNNs can be applied to approximate the items $f_i(Z)$, $i = 1, 2, \dots, n$, respectively, as follows:

$$\begin{aligned} \hat{f}_i(Z) &= \hat{W}_i^T S_i(Z) \\ f_i(Z) &= W_i^{*T} S_i(Z) + \varepsilon_i \end{aligned} \quad (33)$$

where $W_i^{*T} \in R^{N_{0i}}$ and $\hat{W}_i^T \in R^{N_{0i}}$ are the optimal NN weight vector and the estimation of W_i^{*T} , respectively. N_{0i} represents the node number of the i th NN. $S_i(Z) \in R^{N_{0i}}$ represents the basis function vector. The term $\varepsilon_i \in R$ represents the estimation error of the i th NN which is bounded by $\varepsilon_i < \bar{\varepsilon}_i$.

To extend SGUUB of the traditional NNs to GUUB, the following function can be defined:

$$\gamma_i(e_i) = \begin{cases} 1 & |e_i| < d_{1i} \\ \frac{d_{2i}^2 - e_i^2}{d_{2i}^2 - d_{1i}^2} e^{\left(\frac{e_i^2 - d_{1i}^2}{\kappa_i (d_{2i}^2 - d_{1i}^2)} \right)^2} & d_{1i} \leq |e_i| \leq d_{2i} \\ 0 & |e_i| > d_{2i} \end{cases} \quad (34)$$

where d_{1i} and d_{2i} are all positive constants that satisfy $d_{1i} < d_{2i}$. The parameters κ_i are the positive constants with $\kappa_i \geq 1$.

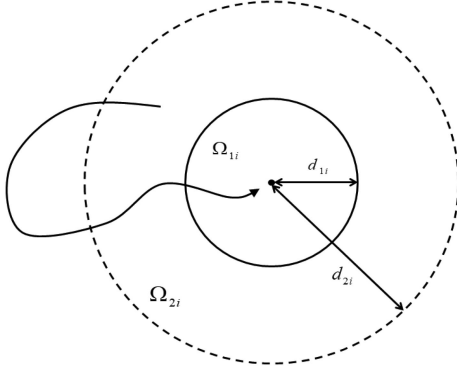


Fig. 2. Global NN tracking performance.

Then the global NN controller can be designed as follows:

$$\begin{aligned} \tau_i = & -k_{4i}z_i - \varsigma_i e_i - \gamma_i(e_i)\hat{f}_i - (1 - \gamma_i(e_i))\bar{f}_i \tanh\left(\frac{z_i \bar{f}_i}{\varpi_i}\right) \\ & - k_{5i}z_i^3 - k_{6i}\text{sig}^{0.5}(z_i) \end{aligned} \quad (35)$$

where k_{4i} , k_{5i} , k_{6i} , and ϖ_i are all proper positive constants satisfying that $k_{4i} > \frac{1}{2}$, $k_{5i} > \frac{1}{2}$. The term $\text{sig}^{0.5}(z_i)$ has the meaning shown in Remark 1.

The adaptive law for the NN weight is designed as

$$\dot{\hat{W}}_i = Q_i \left(\gamma_i z_i S_i(Z) - m_i \hat{W}_i \right) \quad (36)$$

where Q_i is a positive definitive diagonal matrix. The parameter m_i is a positive constant.

Remark 3: From Fig. 2, we can know that when e_i runs in the NN approximation domain Ω_{1i} , the robust item $(1 - \gamma_i(e_i))\bar{f}_i \tanh(\frac{z_i \bar{f}_i}{\varpi_i})$ in (35) turns to 0 and only the item $\gamma_i(e_i)\hat{f}_i$ plays a role. When the state runs out of domain Ω_{2i} , only the robust item plays a role and pulls the state back. When the state is between Ω_{1i} and Ω_{2i} , both the terms play a role and pull the state to the approximation domain Ω_{1i} .

Theorem 1: For the unknown robot manipulator in (1), when the initial conditions are within the output constraints in (11) and (12), the tracking error e_i in (10) can be fixed-time convergent without violating the constraints when the NN controller in (35) and the adaptive law in (36) are employed. Moreover, all the closed-loop signals in the system are GUUB.

Proof: The global Lyapunov function is constructed as follows:

$$V = V_2 + \frac{1}{2} \sum_{i=1}^n \tilde{W}_i^T Q_i^{-1} \tilde{W}_i \quad (37)$$

where $\tilde{W}_i = W_i^* - \hat{W}_i$ can be seen as the estimation error of the optimal NN weight W_i^* .

Taking the derivative of V combined with (32) and (36), we can get

$$\dot{V} = \dot{V}_1 + z^T [\tau + F(Z)] - \sum_{i=1}^n \left(\tilde{W}_i^T z_i \gamma_i S_i(Z) - m_i \tilde{W}_i^T \hat{W}_i \right). \quad (38)$$

Substituting (27), (33), and (35) into (38), we can get

$$\begin{aligned} \dot{V} \leq & \sum_{i=1}^n \left[-\frac{k_{1i}\varphi_i^2}{1-\varphi_i^2} - \frac{k_{2i}(\varphi_i^2)^2}{(1-\varphi_i^2)^2} - \frac{k_{3i}(\varphi_i^2)^{\frac{3}{4}}}{(1-\varphi_i^2)^{\frac{3}{4}}} \right] \\ & + \sum_{i=1}^n \left[-k_{4i}z_i^2 - k_{5i}z_i^4 - k_{6i}(z_i^2)^{\frac{3}{4}} \right] \\ & + \sum_{i=1}^n \left[z_i \gamma_i \varepsilon_i + m_i \tilde{W}_i^T \hat{W}_i \right] \\ & + \sum_{i=1}^n \left[(1 - \gamma_i) \left(f_i z_i - \bar{f}_i z_i \tanh\left(\frac{z_i \bar{f}_i}{\varpi_i}\right) \right) \right]. \end{aligned} \quad (39)$$

From Lemma 3, we can obtain that

$$\begin{aligned} f_i z_i - \bar{f}_i z_i \tanh\left(\frac{z_i \bar{f}_i}{\varpi_i}\right) & \leq \bar{f}_i |z_i| - \bar{f}_i z_i \tanh\left(\frac{z_i \bar{f}_i}{\varpi_i}\right) \\ & \leq c \varpi_i. \end{aligned} \quad (40)$$

And for the terms $\gamma_i z_i \varepsilon_i$ and $m_i \tilde{W}_i^T \hat{W}_i$, we can obtain the following results:

$$\gamma_i z_i \varepsilon_i \leq \gamma_i |z_i| |\varepsilon_i| \leq \frac{1}{2} z_i^2 + \frac{1}{2} \varepsilon_i^2 \leq \frac{1}{2} z_i^2 + \frac{1}{2} \bar{\varepsilon}_i^2 \quad (41)$$

$$\begin{aligned} m_i \tilde{W}_i^T \hat{W}_i & = m_i \tilde{W}_i^T (W_i^* - \tilde{W}_i) = -m_i \|\tilde{W}_i\|^2 + m_i \tilde{W}_i^T W_i^* \\ & \leq m_i \left(\frac{\|W_i^*\|^2}{2} - \frac{\|\tilde{W}_i\|^2}{2} \right) \end{aligned} \quad (42)$$

where $0 \leq \gamma_i \leq 1$ has been used in (41).

Substituting (40), (41), and (42) into (39), we have

$$\begin{aligned} \dot{V} \leq & \sum_{i=1}^n \left[-\frac{k_{1i}\varphi_i^2}{1-\varphi_i^2} - \frac{k_{2i}(\varphi_i^2)^2}{(1-\varphi_i^2)^2} - \frac{k_{3i}(\varphi_i^2)^{\frac{3}{4}}}{(1-\varphi_i^2)^{\frac{3}{4}}} \right] \\ & + \sum_{i=1}^n \left[-k_{4i}z_i^2 - k_{5i}z_i^4 - k_{6i}(z_i^2)^{\frac{3}{4}} \right] + \sum_{i=1}^n \left[\frac{1}{2} z_i^2 + \frac{1}{2} \bar{\varepsilon}_i^2 \right] \\ & + \sum_{i=1}^n \left[m_i \left(\frac{\|W_i^*\|^2}{2} - \frac{\|\tilde{W}_i\|^2}{2} \right) \right] + \sum_{i=1}^n [(1 - \gamma_i) c \varpi_i]. \end{aligned} \quad (43)$$

From (43), we have the following inequality:

$$\begin{aligned} \dot{V} \leq & \sum_{i=1}^n \left[-2k_{1i} \cdot \frac{1}{2} \ln \frac{1}{1-\varphi_i^2} \right] - \min \left(\frac{2k_{4i}-1}{\lambda_{\max}(M)} \right) \frac{1}{2} z^T M z \\ & - \min \left(\frac{m_i}{\lambda_{\max}(Q_i^{-1})} \right) \sum_{i=1}^n \left(\frac{1}{2} \tilde{W}_i^T Q_i^{-1} \tilde{W}_i \right) + \Lambda_1 \end{aligned} \quad (44)$$

where $\frac{\varphi_i^2}{1-\varphi_i^2} \geq \ln \frac{1}{1-\varphi_i^2}$, $\forall |\varphi_i| < 1$ has been used in the derivation of (44). And $\Lambda_1 = \sum_{i=1}^n [\frac{1}{2} \bar{\varepsilon}_i^2 + \frac{m_i \|W_i^*\|^2}{2} + c \varpi_i]$ can be

seen as a bounded positive constant, and $\lambda_{\max}(\cdot)$ is the largest eigenvalue of the term $(\cdot)$.

Then we can obtain the result as follows:

$$\dot{V} \leq -\psi_1 V + \Lambda_1 \quad (45)$$

where $\psi_1 = \min\left(2k_{1i}, \frac{2k_{4i}-1}{\lambda_{\max}(M)}, \frac{m_i}{\lambda_{\max}(Q_i^{-1})}\right)$.

Multiplying both sides by $e^{\psi_1 t}$ in (45) and then integrating both sides over $[0, t]$, we have

$$V \leq \left[V(0) - \frac{\Lambda_1}{\psi_1}\right] e^{-\psi_1 t} + \frac{\Lambda_1}{\psi_1} \leq V(0) + \frac{\Lambda_1}{\psi_1}. \quad (46)$$

Remark 4: From the above inequality, the boundedness of the Lyapunov function $V(t)$ can be inferred. Combined with the definition of $V(t)$ in (17), (29), and (37), the boundedness of $\ln \frac{1}{1-\varphi_i^2}$, z_i , and $\tilde{W}_i$ can be concluded, respectively. So $|\varphi_i| < 1$ can be guaranteed. Then the tracking constraints $\xi_{1i} < e_i < \xi_{2i}$ will not be violated during the operation from (13) and (14). From (10) combined with Assumption 1, we can conclude that q is bounded. Then from the boundedness of φ_i , ξ_{1i} , ξ_{2i} , $\dot{\xi}_{1i}$, and ξ_{2i} , we can conclude the boundedness of the virtual controller α_i . Combined with the boundedness of $\hat{W}_i$ and the Gaussian basis function vector $S_i(Z)$, the controller in (35) can be concluded to be bounded. Therefore, all the closed-loop signals in the system have been proved to be GUUB.

Similar to the derivation of (44), the following inequality can be obtained from (43):

$$\begin{aligned} \dot{V} \leq & \sum_{i=1}^n \left[-\frac{k_{2i}(\varphi_i^2)^2}{(1-\varphi_i^2)^2} - \frac{k_{3i}(\varphi_i^2)^{\frac{3}{4}}}{(1-\varphi_i^2)^{\frac{3}{4}}} - k_{5i}(z_i^2)^2 - k_{6i}(z_i^2)^{\frac{3}{4}} \right. \\ & \left. + \sum_{i=1}^n \left[\frac{1}{2}z_i^2 + \frac{1}{2}\tilde{\varepsilon}_i^2 + \frac{m_i\|W_i^*\|^2}{2} + c\varpi_i \right] \right]. \end{aligned} \quad (47)$$

For the term $\frac{1}{2}z_i^2$, we have

$$\frac{1}{2}z_i^2 \leq \frac{1}{2}z_i^4 + \frac{1}{8}. \quad (48)$$

From Remark 4, we can conclude that $\|\tilde{W}_i\|$ is upper bounded by $\|\tilde{W}_i\| \leq \tilde{\bar{W}}_i$ where $\tilde{\bar{W}}_i$ is defined as an unknown positive constant. Then substituting (48) into (47), we have

$$\begin{aligned} \dot{V} \leq & \sum_{i=1}^n \left[-\frac{k_{2i}(\varphi_i^2)^2}{(1-\varphi_i^2)^2} - \frac{k_{3i}(\varphi_i^2)^{\frac{3}{4}}}{(1-\varphi_i^2)^{\frac{3}{4}}} - \left(k_{5i} - \frac{1}{2}\right)z_i^4 \right. \\ & \left. - k_{6i}(z_i^2)^{\frac{3}{4}} - k_{7i}\left(\|\tilde{W}_i\|^2\right)^2 - k_{8i}\left(\|\tilde{W}_i\|^2\right)^{\frac{3}{4}} \right] + \Lambda_2 \end{aligned} \quad (49)$$

where Λ_2 is a bounded positive constant and $\Lambda_2 = \sum_{i=1}^n [k_{7i}\tilde{\bar{W}}_i^4 + k_{8i}\tilde{\bar{W}}_i^{\frac{3}{4}} + \frac{1}{8} + \frac{1}{2}\tilde{\varepsilon}_i^2 + \frac{m_i\|W_i^*\|^2}{2} + c\varpi_i]$.

Remark 5: For the fixed-time stability analysis which is similar to (49) in [17]–[19], the assumption that $\|\tilde{W}_i\|$ is bounded by $\|\tilde{W}_i\| \leq \tilde{\bar{W}}_i$ was required to integrate the term $k_{7i}(\|\tilde{W}_i\|^2)^2 + k_{8i}(\|\tilde{W}_i\|^2)^{\frac{3}{4}}$ to the positive constant Λ_2 , which lacks theoretical basis. In our work, the Lyapunov function satisfies the inequality like (45). So the boundedness of $\|\tilde{W}_i\|$

can be proved instead of being assumed. Therefore, compared to [17]–[19], the superiority of our work can be reflected.

From (49) and Lemma 2, we can conclude that

$$\dot{V} \leq -\psi_2 V^2 - \psi_3 V^{\frac{3}{4}} + \Lambda_2 \quad (50)$$

where $\frac{\varphi_i^2}{1-\varphi_i^2} \geq \ln \frac{1}{1-\varphi_i^2}$, $\forall |\varphi_i| < 1$ has been used in the above derivation. The term $\psi_2 = \frac{1}{3n} \min[4k_{2i}, \frac{4k_{5i}-2}{\lambda_{\max}^2(M)}, \frac{4k_{7i}}{\lambda_{\max}^2(Q_i^{-1})}]$, $\psi_3 = \min[\frac{k_{3i}}{(\frac{1}{2})^{\frac{3}{4}}}, \frac{k_{6i}}{(\frac{1}{2})^{\frac{3}{4}}\lambda_{\max}^{\frac{3}{4}}(M)}, \frac{k_{8i}}{(\frac{1}{2})^{\frac{3}{4}}\lambda_{\max}^{\frac{3}{4}}(Q_i^{-1})}]$.

So far, the inequality form such as that in Lemma 1 has been obtained. Then the system can be seen as fixed-time convergent independent of the initial conditions. And the system Lyapunov function V will converge to the following compact set in fixed-time:

$$\Omega_g = \left\{ x | V \leq \min \left[\left(\frac{\Lambda_2}{\psi_3(1-\theta_1)} \right)^{\frac{4}{3}}, \left(\frac{\Lambda_2}{\psi_2(1-\theta_1)} \right)^{\frac{1}{2}} \right] \right\} \quad (51)$$

where θ_1 is a positive constant satisfying $\theta_1 \in (0, 1)$. Moreover, the tracking error e_i can be fixed-time convergent and accurately constrained by $\xi_{1i} < e_i < \xi_{2i}$ instead of the compact set in (51) because of the BLF in (17).

Then the convergence time can be represented as

$$T_{g\max} = \frac{4}{\psi_3\theta_1} + \frac{1}{\psi_2\theta_1}. \quad (52)$$

So far, the proof has been completed.

Remark 6: From (46) and (51), we can conclude that the relatively small Λ_1 and Λ_2 and the relatively large ψ_1 , ψ_2 , and ψ_3 will lead to the smaller tracking error without violating the predefined output constraints. From (52), we can conclude that relatively large ψ_2 and ψ_3 will lead to shorter convergence time. From the definitions of the above terms in the article, we can see that the relatively small Λ_1 and Λ_2 can be obtained from the relatively small ϖ_i and m_i . The relatively large ψ_1 , ψ_2 , and ψ_3 can be obtained from the relatively large m_i , Q_i , and k_{ji} ($j = 1, 2, \dots, 6$). The above theoretical analysis can guide the parameter adjustment combined with the operator's experience to obtain better control performance.

IV. SIMULATION STUDY

For the sake of simplicity, a 2-DOF robot will be employed to illustrate the effectiveness of the proposed controller.

A. Simulation Setup

When $n = 2$ in (1), the terms $M(q)$, $C(q, \dot{q})$, and $G(q)$ can be, respectively, defined as follows:

$$\begin{aligned} M(q) &= \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \\ C(q, \dot{q}) &= \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \\ G(q) &= \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} \end{aligned} \quad (53)$$



Fig. 3. Schematic diagram of Baxter robot for experiment.

where

$$\begin{aligned}
 M_{11} &= m_1 l_{c1}^2 + m_2 (l_1^2 + l_{c2}^2 + 2l_1 l_{c2} \cos q_2) + I_1 + I_2 \\
 M_{12} &= M_{21} = m_2 (l_{c2}^2 + l_1 l_{c2} \cos q_2) + I_2 \\
 M_{22} &= m_2 l_{c2}^2 + I_2 \\
 C_{11} &= -m_2 l_1 l_{c2} \dot{q}_2 \sin q_2 \\
 C_{12} &= -m_2 l_1 l_{c2} (\dot{q}_1 + \dot{q}_2) \sin q_2 \\
 C_{21} &= m_2 l_1 l_{c2} \dot{q}_1 \sin q_2 \\
 C_{22} &= 0 \\
 G_1 &= (m_1 l_{c2} + m_2 l_1) g \cos q_1 + m_2 l_{c2} g \cos(q_1 + q_2) \\
 G_2 &= m_2 l_{c2} g \cos(q_1 + q_2)
 \end{aligned} \tag{54}$$

with $m_1 = m_2 = 2 \text{ kg}$ and $l_1 = l_2 = 0.5 \text{ m}$ being the mass and the length of the i th ($i = 1, 2$) link, respectively. And l_{ci} ($i = 1, 2$) denotes the distance between the $i - 1$ th joint and the i th link's mass center. Then $I_1 = I_2 = 0.125 \text{ kgm}^2$ can be calculated as the inertia of the i th link.

The desired trajectory is set to $q_d = [\pi/2 + 0.7\cos(2t) - 0.7, \pi/4 + 0.8\sin(2t)]^T$. In order to demonstrate the fixed-time convergence of the proposed controller, we set the initial errors of the two links as $[0.1, -0.1]^T$, $[0.2, -0.2]^T$, $[0.4, -0.4]^T$, and $[0.8, -0.8]^T$, respectively. The output constraints ξ_{1i} and ξ_{2i} are set as

$$\begin{aligned}
 \xi_{11} &= \xi_{12} = -[(1 - 0.03)e^{-t} + 0.03] \\
 \xi_{21} &= \xi_{22} = (1 - 0.05)e^{-t} + 0.05.
 \end{aligned} \tag{55}$$

For the controller design, the parameters can be set as: $k_{1i} = k_{2i} = k_{3i} = 2$, $k_{a1} = k_{a2} = 0.5$, $k_{4i} = k_{5i} = k_{6i} = 2$, and $\varpi_i = 0.01$. The boundaries of the NN approximation term

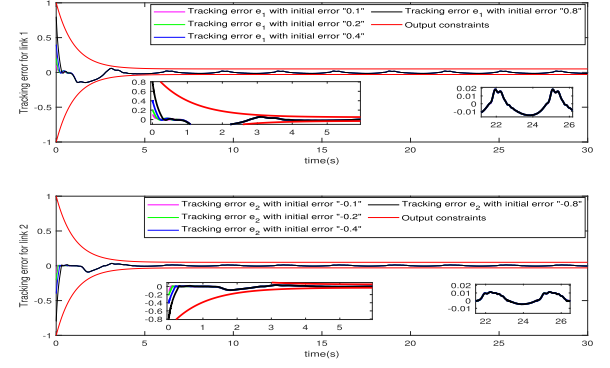


Fig. 4. Tracking errors of 2-DOF robot manipulator with the proposed controller.

and the robust term are set as $d_{1i} = 0.1$ and $d_{2i} = 1$ and the parameter $\varkappa_i$ is set as $\varkappa_i = 1.2$. The parameters for the adaptive law in (36) are set as $Q_1 = \text{diag}\{0.18\}$, $Q_2 = \text{diag}\{0.4\}$ and $m_1 = 1.7$, $m_2 = 0.5$.

In addition, to further verify the superiority of our work, an adaptive NN controller in [32] has been applied to the same 2-DOF robot model in the article. To be fair, all the physical parameters, the predefined output constraints, and the desired trajectory of the robot system are set the same as that in the above simulation study. And the initial errors are set the same as $[0.1, -0.1]^T$ with the two different controllers. By the repeated simulation to achieve better control performance, for the following controlled terms in [32]

$$\alpha_{1i} = \dot{y}_{di} - k_{1i} e_{1i} \tag{56}$$

$$u_i = -k_{2i} e_{2i} - \frac{1}{2a_i^2} e_{2i} \hat{\theta}_i \phi^T \phi \tag{57}$$

$$\dot{\hat{\theta}}_i = \frac{\lambda_i}{2a_i^2} e_{2i} \phi^T \phi - \gamma_i \hat{\theta}_i \tag{58}$$

the parameters are, respectively, designed as $k_1 = [30, 15]^T$, $k_2 = [19.6, 9.6]^T$, $a_i = 1.5$, and $\gamma_i = 1$ for $i = 1, 2$.

B. Simulation Results

The simulation results have been shown as Figs. 4–6. From Figs. 4 and 5, we can see that both e_i and $\hat{W}_i$ can achieve fixed-time convergence with different initial conditions, while e_i can be convergent to $[-0.02, 0.02]$ in less than 5 s without violating the output constraints. The comparison simulation results have been shown in Fig. 6. We can see that quite a good tracking performance can be obtained using the adaptive NN controller in [32]. In fact, better control performance can be obtained with more proper designed parameters in [32] during the process of our massive repeated simulation. However, the predefined output constraints may be violated with the controller in [32] while the suboptimal parameters are being selected, such as the results in Fig. 6. But the same constraints can always be handled in our work. So the superiority of the proposed control scheme has been shown in the aspect of the transient performance.

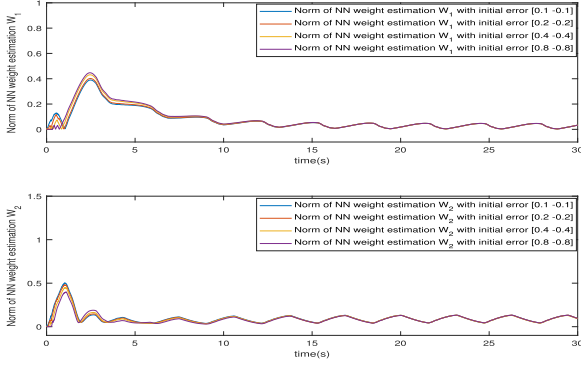


Fig. 5. NN weight estimation of 2-DOF robot manipulator with proposed controller.

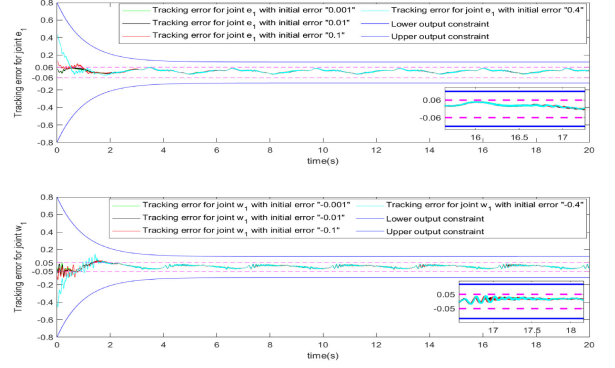


Fig. 7. Tracking errors of left e_1 and w_1 joints with the proposed controller.

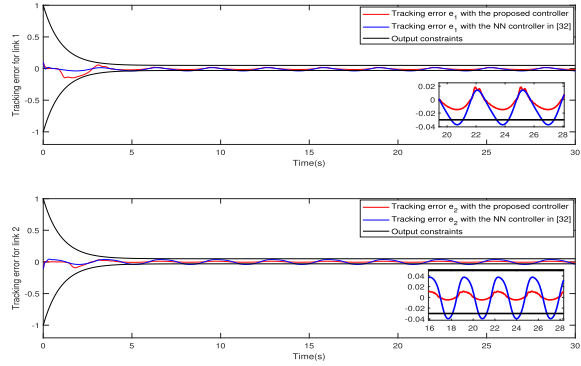


Fig. 6. Tracking errors of 2-DOF robot manipulator with different controller.

V. EXPERIMENT STUDY

To verify the practicability of the proposed control scheme, a Baxter robot, as shown in Fig. 3, is employed in the experiment. It is a bimanual robot with advanced sensing technologies and control at every joint of its two 7-DOF arms. The detailed information of the kinematics and dynamics of the Baxter robot has been given in [33] and [34]. For simplicity and without loss of generality, we select two joints, i.e., left e_1 and left w_1 of the left arm to derive the motion for experiment.

Remark 7: From (21), we can see when the value of the tracking errors are closer to the limits of the output constraints ξ_{1i} and ξ_{2i} , the values of the terms $1 - \varphi_{ai}$ and $1 - \varphi_{bi}$ will be closer to zero, which will lead to very large α_i because of their role of denominator. Then the large α_i may lead to large torques in (35) that exceed the capacity of the machine. So we set the slightly larger values of $\phi_{\infty,1i}$ and $\phi_{\infty,2i}$ in (11) and (12). Similarly, we set the slightly smaller initial conditions in case the initial torques are too large to cause the safety problem.

A. Experiment Setup

The desired trajectory is set to $q_d = [\pi/2 + 0.7\cos(2t) - 0.7, \pi/4 + 0.8\sin(2t)]^T$. In order to demonstrate the fixed-time convergence of the proposed controller, we set the initial errors of left e_1 and left w_1 as $[0.001, -0.001]^T$, $[0.01, -0.01]^T$,

$[0.1, -0.1]^T$, and $[0.4, -0.4]^T$, respectively. The output constraints ξ_{1i} and ξ_{2i} are set as

$$\begin{aligned}\xi_{11} &= \xi_{12} = -[(0.8 - 0.12)e^{-t} + 0.12] \\ \xi_{21} &= \xi_{22} = (0.8 - 0.12)e^{-t} + 0.12.\end{aligned}\quad (59)$$

For the controller design, the parameters can be set as $k_{j1} = 1.8$, $k_{j2} = 0.8$ ($j = 1, 2, \dots, 6$), $k_{a1} = 0.5$, $k_{a2} = 0.4$, and $\varpi_i = 0.01$. The boundaries of the NN approximation term and the robust term are set as $d_{1i} = 0.2$ and $d_{2i} = 0.5$ and the parameter $\varkappa_i = 1.2$. The parameters for the adaptive law in (36) are set as $Q_1 = \text{diag}\{0.03\}$, $Q_2 = \text{diag}\{0.02\}$ and $m_1 = 2$, $m_2 = 15$.

To further verify the superiority of the proposed NN control scheme, the NN controller in [32] and a PD controller have been applied to the same Baxter robot in the article. To be fair, the predefined output constraints and the desired trajectory of the robot system are set the same as that in the above experiment study. And the initial errors are set the same as $[0.1, -0.1]^T$ with the three different controllers. By the repeated experiment study to achieve better control performance, we set the parameters in (56), (57), and (58) as $k_1 = [13, 4]^T$, $k_2 = [0.5, 0.5]^T$, $a_i = 1.5$, and $\gamma_i = 1$ for $i = 1, 2$, respectively. And the PD controller is designed as $\tau = -K_p e - K_d \dot{e}$ where $K_p = \text{diag}(50, 40)$ and $K_d = \text{diag}(\sqrt{0.5}, 0.5)$.

B. Experimental Results

The experimental results have been shown in Figs. 7–9. From Figs. 7 and 8, we can see that both e_i and $\hat{W}_i$ can achieve fixed-time convergence with different initial conditions, while e_i can be, respectively, convergent to $[-0.06, 0.06]$ and $[-0.05, 0.05]$ in less than 3 s without violating the output constraints. The comparison experimental results have been shown in Fig. 9. We can see that quite a good tracking performance can be obtained using both the adaptive NN controller in [32] and the PD controller. In fact, better control performance can be obtained with more proper designed parameters in the two comparative controllers during the process of our massive repeated experiment. However, the predefined output constraints may be violated in the comparison experiment while the suboptimal

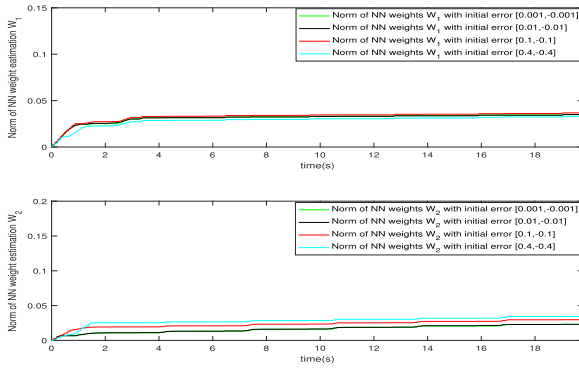


Fig. 8. NN weight estimation of W_1 and W_2 with the proposed controller.

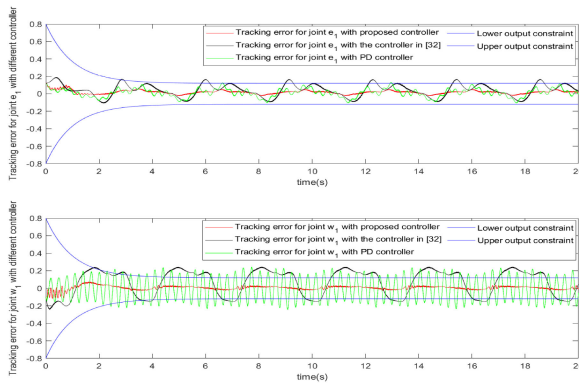


Fig. 9. Tracking errors of left e_1 and w_1 joints with different controllers.

parameters are being selected, such as the results in Fig. 9. But the same constraints can always be handled in our work. So the superiority of the proposed control scheme has been shown in the aspect of the transient performance.

Remark 8: Although the above superior control performance with the proposed controller is exciting, the defect that exists for the convergence time in (52) cannot be accurately calculated because of the unknown parameter θ_1 , which has been the common problem of the practical fixed-time control schemes. But similar to the existing fixed-time control schemes, the convergence time is very nearly the same with different initial conditions, which has been shown in the simulation and experiment study. So the superiority of the proposed control scheme has been shown in the aspect of the convergence time.

VI. CONCLUSION

In this article, a global fixed-time NN controller was proposed for unknown robot manipulators with predefined output constraints. A switching mechanism was integrated into the controller design such that SGUUB in the traditional NN control schemes can be extended to GUUB. The system Lyapunov function satisfies two different forms of inequalities like (45) and (50) simultaneously such that the assumption that the NN weight estimation is bounded can be relaxed compared with the existing fixed-time NN control schemes [17]–[19]. A BLF was

exploited such that the tracking errors can be constrained in the predefined range, rather than the uncertain convergence domain in most practical fixed-time algorithms [17], [18], [27]. On the whole, compared with the traditional NN control algorithms, the proposed NN controller in this article can guarantee the global stability and the fixed-time convergence simultaneously without violating the predefined output constraints, which showed the superiority of the control scheme.

In the future work, the neural control problem of unknown bimanual robots with fixed-time convergence will be discussed.

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Chengzhi Zhu received the B.S. degree in automation from Qinghai University, Xining, China, in 2018. He is currently working toward the Ph.D. degree majoring in control science and engineering with the South China University of Technology, Guangzhou, China.

He was an exchange student with the East China University of Science and Technology, Shanghai, China, from 2015 to 2016. His current research interests include robotics, intelligent control, and human–robot interaction.



Yiming Jiang received the B.S. degree in automation from Hunan University, Changsha, China, in 2011, the M.S. degree in control theory and engineering from the School of Automation, Guangdong University of Technology, Guangzhou, China, in 2015, and the Ph.D. degree from the School of Control Science and Engineering, South China University of Technology, Guangzhou, in 2019.

He was a Visiting Scholar with the University of Portsmouth, Portsmouth, U.K., from 2017 to 2018. He is currently an Associate Professor with the School of Robotics, Hunan University. His research interests include robotics, intelligent control, and human–robot interaction.



Chenguang Yang (Senior Member, IEEE) received the B.Eng. degree in measurement and control from Northwestern Polytechnical University, Xi'an, China, in 2005, the Ph.D. degree in control engineering from the National University of Singapore, Singapore, in 2010, and completed the Postdoctoral Training in human robotics from the Imperial College London, London, U.K., in 2010.

His research interests include robotics and automation.

Dr. Yang was a recipient of the IEEE Transactions on Robotics Best Paper Award in 2012 and the IEEE Transactions on Neural Networks and Learning Systems Outstanding Paper Award in 2022 as the lead author.